

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1652

Lawnwood 138 kV
March 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1652. The Interconnection Customer proposes a 122.3 MW solar photovoltaic facility interconnecting at the Lawnwood 138 kV substation in the Pennsylvania area. The request was received on January 21, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 557 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$43,432,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Montour Sta – Talenenergy Martins Crk Llc Williamsport 345.0 kV Line, the E Fayetteville – El Dorado 138.0 kV loads to 187.5% of its 192 MVA rating. Project DFAX: 18.4%.

The El Dorado – Lilac 138.0 kV is overloaded to 180.5% of its 193 MVA emergency rating for the loss of the Montour Sta – Talenenergy Martins Crk Llc Williamsport 345.0 kV. The project contributes 18.4% DFAX (22.5 MW) to this overload.

For the loss of Montour Station – Talenenergy Martins Creek Llc Williamsport 345.0 kV, the East Fayetteville – Coopervision 138.0 kV loads to 165.0% of its 232 MVA rating. Project DFAX: 18.4%.

Johnson – Lancaster Substation 138.0 kV Line: 132.3% loading (185 MVA) following the outage of the Montour Sta – Acadia 500.0 kV; DFAX = 5.7%.

For the loss of Talenenergy Martins Crk Llc Williamsport – Montour Station 345.0 kV L, the Coopervision – Power Inn 138.0 kV loads to 125.2% of its 225 MVA rating. Project DFAX: 8.6%.

The Lawton E Side – Johnson 138.0 kV is overloaded to 120.3% of its 196 MVA emergency rating for the loss of the Acadia – Montour Station 500.0 kV. The project contributes 5.7% DFAX (7.0 MW) to this overload.

For the loss of Montour Station – Talenenergy Martins Crk Llc Williamsport 345.0 kV, the Talenenergy Martins Creek Llc Williamsport – John Day 2 138.0 kV loads to 116.4% of its 297 MVA rating. Project DFAX: 6.3%.

For the loss of Montour Sta – Talenenergy Martins Creek Llc Williamsport 345.0 kV, the Montour Station – Lilac 138.0 kV line loads to 104.7% of its 246 MVA rating. Project DFAX: 14.5%.

Panda Patriot Generation Plant – Finley Butte 138.0 kV: 104.2% loading (185 MVA) following the outage of the East Fayetteville – Coopervision 138.0 kV; DFAX = 5.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Fayetteville – El Dorado 138.0 kV	Rebuild line with higher-capacity conductor	\$4,364,000	100% (\$4,364,000)
El Dorado – Lilac 138.0 kV	Rebuild line with higher-capacity conductor	\$6,455,000	100% (\$6,455,000)
East Fayetteville – Coopervision 138.0 kV	Rebuild line with higher-capacity conductor	\$7,426,000	100% (\$7,426,000)
Johnson – Lancaster Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$20,138,000	57% (\$11,557,000)
Coopervision – Pwer Inn 138.0 kV	Rebuild line with higher-capacity conductor	\$7,111,000	100% (\$7,111,000)
Johnson – Lawton East Side 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,553,000	57% (\$2,039,000)
Talenenergy Martins Creek Llc Williamsport – John Day 2 138.0 kV	Rebuild line with higher-capacity conductor	\$9,249,000	16% (\$1,485,000)
Lilac Substation – Montour Ta 138.0 kV	Reconductor line	\$1,112,000	100% (\$1,112,000)
Panda Patriot Generation Plant – Finley Butte 138.0 kV	Reconductor line	\$7,385,000	26% (\$1,883,000)
	Total allocated to project		\$43,432,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.